CSE 717 — Information Security

Topic 6: Markov Chains — Steady-State Probability, 2-State Security Model

Complete Past Paper Solutions: 2020–2024

Source: Lc#4A (Markov Chain theory) + Lc#4B (Assignment numericals)

2/5 years (2023, 2024 — both most recent), 3–4 marks. Strong recency signal — near-certain repeat.

Contents

Quick Reference — Steady-State Method (CHEAT SHEET)

2020–2022: no Markov Chain question found

This topic first appeared in 2023 and 2024 (both most recent years) as a 2-state security model: **Secure/Compromised** or **Secure/Insecure**. The method is identical both years — only the transition probabilities change. Master one worked example cold.

Key Definitions (Lc#4A)

- **Markov chain:** a model where the next state depends *only* on the current state (memoryless / Markov property).
- **State:** a possible situation of the system (e.g. Secure, Compromised, Sunny, Rainy).
- **Transition probability:** the probability of moving from one state to another in one step. All outgoing probabilities from a state must sum to 1.
- **Transition matrix T:** $k \times k$ matrix for k states; entry T_{ij} = probability of going from state i to state j. Each *row* sums to 1.
- **State vector $x^{(n)}$:** a row vector of probabilities of being in each state at step n. Updated as $x^{(n)} = x^{(n-1)}T$.
- **Steady-state vector V:** the long-run probability distribution, satisfying $VT = V$ (does not change after further steps). Also written $V(T - I) = 0$ with the constraint $\sum V_i = 1$.

5-Step Algorithm to Find Steady-State (Lc#4A standard method)

1. **Set up transition matrix T** from the given probabilities (rows = "from", columns = "to").
2. **Write $V(T - I) = 0$:** subtract the identity matrix I from T, multiply row

vector $V = [S \;\; C]$ on the left, set equal to zero row vector.

3. **Extract equations**: one equation per column (they are linearly dependent — any one can be dropped). This gives a relation between S and C.
4. **Use the normalization constraint** $S + C = 1$ (probabilities must sum to 1) together with the relation from Step 3.
5. **Solve for** S **and** C, state the steady-state vector $V = [S \;\; C]$, and **interpret** (which state dominates long-run?).

Worked example from slides (Lc#4A) — Sunny/Rainy weather

$$T = \begin{pmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{pmatrix} \quad \text{(Sunny→Sunny=0.9, Rainy→Sunny=0.5)}$$

$$T - I = \begin{pmatrix} -0.1 & 0.1 \\ 0.5 & -0.5 \end{pmatrix}$$

$$[S \;\; R] \begin{pmatrix} -0.1 & 0.1 \\ 0.5 & -0.5 \end{pmatrix} = [0 \;\; 0] \;\Rightarrow\; -0.1S + 0.5R = 0 \;\Rightarrow\; S = 5R$$

With $S + R = 1$: $5R + R = 1 \Rightarrow R = 1/6 \approx 0.167$, $S = 5/6 \approx 0.833$.

$\boxed{V = [0.833 \;\; 0.167]}$ — long-run: 83.3% sunny days.

2024 — Q1(c): Secure/Compromised 2-state system — find steady-state, interpret (3)

Question — 2024 Q1(c) [3 marks]

An information system can be in one of two states: **Secure (S)** or **Compromised (C)**.

- If the system is **Secure** during a security assessment cycle (active patching, effective controls), it remains Secure with probability **0.4**. Otherwise it transitions to Compromised.
- If the system is **Compromised** during an assessment cycle (failed controls, unpatched vulnerabilities), incident-response actions return it to Secure with probability **0.3**. Otherwise it remains Compromised.

Determine the steady-state probabilities of Secure (S) and Compromised (C) states, and evaluate whether the system is more likely to be secure or compromised in the long run.

Answer — 2024 Q1(c)

Step 1 — Transition Matrix.
From the given probabilities:

- Secure → Secure: 0.4; Secure → Compromised: $1 - 0.4 = 0.6$
- Compromised → Secure: 0.3; Compromised → Compromised: $1 - 0.3 = 0.7$

$$T = \begin{pmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{pmatrix} \quad \text{(rows = “from” state, columns = “to” state)}$$

Step 2 — Set up steady-state equation $V(T-I)=0$.

$$T - I = \begin{pmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -0.6 & 0.6 \\ 0.3 & -0.3 \end{pmatrix}$$

$$[S \;\; C] \begin{pmatrix} -0.6 & 0.6 \\ 0.3 & -0.3 \end{pmatrix} = [0 \;\; 0]$$

Step 3 — Extract equation from column 1.

$$-0.6S + 0.3C = 0 \;\Rightarrow\; C = 2S \quad \cdots (1)$$

(Column 2 gives $0.6S - 0.3C = 0$ — the same equation, linearly dependent. Discard it.)

Step 4 — Apply normalization constraint $S + C = 1$.

$$S + C = 1 \quad \cdots (2)$$

Step 5 — Solve.
Substituting (1) into (2):

$$S + 2S = 1 \;\Rightarrow\; 3S = 1 \;\Rightarrow\; \boxed{S = \tfrac{1}{3} \approx 0.333}$$

$$C = 2S = \tfrac{2}{3} \;\Rightarrow\; \boxed{C \approx 0.667}$$

Steady-state vector:

$$V = [S \;\; C] = [0.333 \;\; 0.667]$$

Interpretation: In the long run, the system is in the **Secure** state only **33.3%** of the time and in the **Compromised** state **66.7%** of the time.

$$\boxed{\text{The system is more likely COMPROMISED in the long run (66.7\% vs 33.3\%).}}$$

Intuition: although incident-response recovers from Compromised state 30% of the time, the patching mechanism only keeps the system Secure 40% of the time — so the system drifts towards compromise.

2023 — Section B Q6(c): Secure/Insecure network with patch-cycle transitions — long-term behaviour (4)

Question — 2023 Section B Q6(c) [4 marks]

Consider a network system that can either be in a **secure state** or an **insecure state**. The network system periodically undergoes patches to maintain security. However, there is a risk that the system may be vulnerable to attacks at any given time.

(i) When the system remains secure because the patching mechanism is effective and there are no new vulnerabilities, there is an **80% chance** that the system remains secure after a security check.

(ii) When no patch is applied or the patching mechanism fails, the system remains insecure and there is a **60% chance** that the system remains secure after a security check.

Determine the long-term behaviour of this network system, whether it is secure or insecure.

Answer — 2023 Section B Q6(c)

Step 1 — Transition Matrix.
From the given probabilities:

- Secure $\to$ Secure: 0.8; Secure $\to$ Insecure: $1 - 0.8 = 0.2$
- Insecure $\to$ Secure: 0.6; Insecure $\to$ Insecure: $1 - 0.6 = 0.4$

$$T = \begin{pmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{pmatrix} \quad \text{(rows = "from", columns = "to")}$$

Step 2 — Set up steady-state equation $V(T - I) = 0$.

$$T - I = \begin{pmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -0.2 & 0.2 \\ 0.6 & -0.6 \end{pmatrix}$$

$$[S \;\; I] \begin{pmatrix} -0.2 & 0.2 \\ 0.6 & -0.6 \end{pmatrix} = [0 \;\; 0]$$

Step 3 — Extract equation from column 1.

$$-0.2S + 0.6I = 0 \Rightarrow S = 3I \quad \cdots (1)$$

(Column 2 gives $0.2S - 0.6I = 0$ — same equation. Discard.)

Step 4 — Apply normalization constraint $S + I = 1$.

$$S + I = 1 \quad \cdots (2)$$

Step 5 — Solve.
Substituting (1) into (2):

$$3I + I = 1 \Rightarrow 4I = 1 \Rightarrow \boxed{I = 0.25}$$

$$S = 3I = 3 \times 0.25 \;\Rightarrow\; \boxed{S = 0.75}$$

Steady-state vector:

$$V = [S \;\; I] = [0.75 \;\; 0.25]$$

Interpretation: In the long run, the system is in the **Secure** state **75%** of the time and in the **Insecure** state only **25%** of the time.

$$\boxed{\text{The system is more likely SECURE in the long run (75\% vs 25\%).}}$$

Intuition: even when insecure, the 60% patch-recovery rate is strong. Combined with 80% staying-secure rate, the patching mechanism is effective overall — the system gravitates toward the secure state.

2020–2022 — No Markov Chain question found

2020/2021/2022 papers

No Markov Chain question was found in the 2020, 2021, or 2022 past papers for CSE-717 / CSE-7197. The topic first appears in 2023 and 2024 (consecutive most recent years) — a strong signal that it has been added to the repeating question pool and will very likely appear in 2026.

Exam Strategy Summary

High-Yield Checklist for Markov Chains (2/5 years, both most recent, 3–4 marks)

1. **Transition matrix orientation:** rows = "from" state, columns = "to" state. Each **row sums to 1**. Check this immediately after writing the matrix.
2. **Fill in the missing probability:** if "Secure→Secure = 0.4" is given, then "Secure→Compromised = 0.6" automatically. Never leave a row incomplete.
3. **Steady-state equation:** $VT = V \;\Leftrightarrow\; V(T - I) = 0$. Expand to get one useful linear equation (the two equations are always dependent — keep only one).
4. **Normalization:** $\pi_1 + \pi_2 = 1$ is the second equation. Together with the linear relation, solve in two lines.
5. **Always interpret the result:** state which system (Secure vs Compromised/Insecure) dominates, with the exact long-run percentage. This interpretation is the final mark.
6. **Both exam questions are 2-state security models** (Secure/Compromised in 2024, Secure/Insecure in 2023). Expect the same framing in 2026 — the lecture worked example is a weather chain (Sunny/Rainy), but map the method directly onto the security context.

Key contrast between the two years:

	2024	**2023**
States	Secure / Compromised	Secure / Insecure
Stay-in-Secure prob	0.4	0.8
Recover-to-Secure prob	0.3	0.6
π_{Secure}	$1/3 \approx 0.333$	$3/4 = 0.75$
$\pi_{\text{Compromised/Insecure}}$	$2/3 \approx 0.667$	$1/4 = 0.25$
Long-run verdict	Compromised	Secure